INCH -POUND
MIL-R-22097G
AMENDMENT 3
31March 1996
SUPERSEDING
AMENDMENT 2
21 April 1995

MILITARY SPECIFICATION

RESISTOR, VARIABLE, NONWIRE-WOUND
(ADJUSTMENT TYPE),
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-22097G, dated 26 November 1993, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 5, delete "120°C" and substitute "125°C".

PAGE 2

TABLE I, note 2, delete and substitute:

"2. For characteristic C only, contact resistance variation shall not exceed 3 percent or 20 ohms, whichever is greater. For characteristic F only, the contact resistance variation shall not exceed 3 percent or 3 ohms, whichever is greater."

PAGE 3

2.1.1, SPECIFICATIONS, MILITARY: Delete "MIL-C-45662 - Calibration Systems Requirements."

PAGE 4

*Add the following paragraph:

"2.1.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the Issues of the documents cited in the solicitation (see 6.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NCSL Z540-1 - General requirements for Calibration Laboratories and Measuring and Test Equipment.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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INTERNATIONAL ORGANIZATION FOR STANDRADS (ISO)

ISO 10012-1 - Quality Assurance Requirements for Measuring Equipment, Part 1: Meteorological Confirmation System For Measuring Equipment.

(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036)."

PAGE 7

3.6, delete "$V = \sqrt{PR}$" and substitute "$E = \sqrt{PR}$".

PAGE 8

3.20, delete "Resistance between end terminals, with the contact arm in the position for setting", and substitute "Resistance between end terminals, with the contact arm in the position for setting stability".

PAGE 9

TABLE V, column 2, row 2, title, delete "%°C" and substitute "PPM/°C".

PAGE 10

3.30.3, last line, delete "removing" and substitute "removed".

4.1.1, line 7, delete "eigher" and substitute "either".

*4.1.2, last sentence, delete "MIL-C-45662" and substitute "ANSI/NCSL Z540-1 ISO 10012-1 or equivalent system, as approved by the qualifying activity."

PAGE 11

4.4.4b, delete in its entirety.

Paragraph following 4.4.4b, second sentence, delete "In addition to the periodic submission of inspection data,".

PAGE 15

4.5.2 through 4.5.2.1.1.1 inclusive, delete in their entirety.

PAGE 16

TABLE XI, delete in its entirety.

4.5.2.1.1.2 through 4.5.2.1.3 inclusive, delete in their entirety.

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PAGE 18

*4.6.4 , delete in its entirety and substitute the following:

"4.6.4 Contact resistance variation (see 3.9). Contact resistance variation shall be measured with the measuring circuit shown on figure 3, or its equivalent. The operating shaft shall be rotated in both directions through 90 percent of the actual effective electrical travel for a maximum of six cycles. Only the last 3 cycles shall count in determining whether or not a contact resistance variation is observed at least twice in the same location, exclusive of the roll-on or roll-off points where the contact arm moves from the termination, on or off, the resistance element. Group A, subgroup I product acceptance may be determined based on one cycle minimum where compliance to the specification is demonstrated. The rate of rotation of the operating shaft shall be such that the wiper completes one cycle in 5 seconds minimum, to 2 minutes maximum."

PAGE 19

4.6.6, line 1, delete "301" and substitute "302".

4.6.7.2, at the end of the paragraph, add "(see 3.12.2)".

4.6.8c, at the end of the sentence, add "(see 3.13)".

PAGE 21

4.6.10d(2), delete "For group B inspection: 10 cycles" and substitute "For group C inspection: 10 cycles."

PAGE 23

* 4.6.16.2, delete in its entirety and substitute:

"4.6.16.2 Procedure. Total resistance shall be measured as specified in 4.6.2.1 and 4.6.8.1, respectively. The resistor shall be placed in a chamber at room temperature. The temperature shall be gradually decreased to -55°C +0°C, -5°C within a period off not less than 1.5 hours. For quality conformance inspection only, and at the option of the manufacturer, the resistors may be placed in the chamber when the chamber is already at the extreme low temperature. After 1 hour of stabilization at this temperature, setting stability shall be measured as specified 4.6.8.1. Full rated continuous working voltage (see 3.1 and 3.6) shall be applied for 45 minutes. The resistors may be loaded individually or in parallel. Fifteen +5, -0 minutes after removal of voltage, setting stability shall be measured as specified 4.6.8.1. The temperature in the chamber shall be gradually increased to room temperature within a period of not more than 8 hours. The resistors shall be removed from the chamber, and maintained at a temperature of 25°C ±5°C for a period of approximately 24 hours. Total resistance shall be measured as specified 4.6.2.1. Resistors shall then be examined for evidence of mechanical damage."

PAGE 28

TABLE XV, column 4, title, delete "Tests from table IV to be performed on additional sample units" and substitute "Tests from table VI to be performed on additional sample units".

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The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate requirements of this document based on the entire content irrespectively of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Preparing activity:
DLA - ES

(Project 5905-1437)

Review activities:
Army - AR, AT, AV, ME, MI
Navy - AS, MC, OS
Air Force - 17, 19, 99

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